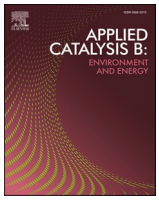
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Plasma catalytic hydrogenolysis of polyethylene to gasoline-range fuels over Sm/USY catalysts

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ABSTRACT

Plasma catalysis, as a highly promising technology for plastic upcycling, enables the efficient conversion of low-density polyethylene (LDPE) into oil fuels under mild conditions. For the plasma-catalytic LDPE hydrogenolysis process at an input power of ca. 56 W without external heating, the optimized 5Sm/USY catalyst exhibited the highest yield of oil fuels (83 %), with C_4-C_{13} gasoline-range fractions accounting for up to 98 %. Compared to pure plasma condition, the incorporation of catalyst improved the discharge power and the discharge mode was transformed to surface discharge, which promoted the LDPE conversion. Notably, the plasma-catalytic system exhibited a 1.51-fold increase in C-C bond cleavage rate compared to thermal catalysis, with energy consumption being less than one-fifth of thermal catalytic processes, demonstrating its superior bond-breaking efficiency and energy-saving advantage. Mechanistic studies revealed that the plasma effectively activates LDPE molecular chains, synergizing with the shape-selective effect of porous USY zeolite. The addition of Sm_2O_3 species increased the content of Lewis acid sites and further promoted hydrogen ion abstraction to form carbocations. This method demonstrated excellent performance in converting other polyolefin wastes and real plastic wastes, particularly achieving 88 % liquid yield from real plastics, fully confirming the exceptional performance of the 5Sm/USY catalyst in plasma-assisted plastic waste conversion.

1. Introduction

Currently, plastic materials have been extensively utilized in agriculture, construction, packaging, and manufacturing industries due to their low cost, multifunctionality and ease of processing, significantly enhancing daily life convenience [1–3]. Nevertheless, the majority of plastic products exhibit a limited functional lifespan, and some inherent drawbacks such as slow decomposition rates and difficulties in recycling, which further resulted in massive accumulation of plastic wastes, inflicting tremendous damage on ecological systems and endangering public health [4–6]. According to the Organization for Economic Co-operation and Development (OECD), the amount of global plastic waste recycling was estimated to be only 9 % as of 2019 [7]. Meanwhile, with plastic products increased, it is predicted to rise to more than 12,000 metric tons of plastic wastes by 2050 [8]. Wherein, polyolefins such as polyethylene (PE) and polypropylene (PP) are predominant (ca. 60 %) in plastic wastes [9–11]. In response to these challenges in ecosystem, chemical recycling of plastic wastes with rich hydrogen and carbon sources has received considerable attention to produce chemicals and fuels, especially for catalytic recycling.

Typically, in the chemical recycling methods, catalytic recycling of plastic wastes could directly transform into monomers or value-added fuels via the directly catalytic role of the specific catalysts [12–14]. For example, Lim et al. achieved 74.8 % low-density polyethylene (LDPE) conversion and 43.8 % liquid yield over UiO-66 catalyst at 400 °C for 5 h [15]. Wang et al. synthesized alloyed $RuPt/ZrO_2$ catalyst for PE hydrogenolysis, achieving 93.9 % conversion to C_1-C_{19} products at 240 °C and 3 MPa H_2 conditions [16]. Wang et al. developed Ru_1/ZrO_2 catalyst by the atomic Ru doped ZrO_2, achieving complete PP conversion via hydrocracking within 4 h at 300 °C under 3 MPa H_2, with 75 % liquid yield with a carbon number range of C_4-C_{17} [17]. Han et al. achieved 88.4 % PE conversion over SO_4/ZrO_2-Al_2O_3 catalyst through a one-pot method under 0.1 MPa N_2 pressure at 280 °C for 48 h, yielding 73.3 % gasoline range hydrocarbons (C_4-C_{13}) [18]. Owing to the inert

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C-C and C-H bonds in polyolefin wastes, traditionally catalytic degradation of them still requires operated at harsh conditions including high temperatures and pressures, prolong reaction time, which leads to high energy consumption and also makes it commercially unattractive [19,20]. In this regard, seeking some technologies that enable effectively transform the electric and/or photonic energies into the driving forces toward activating chemical bonds to achieve plastic wastes depolymerization under relatively mild conditions, is necessary to minimize energy costs [21–23].

Alternatively, plasma-catalytic recycling of polyolefin wastes is a promising route for effectively cleave C-C bonds and C-H bonds under milder conditions than traditionally thermal-catalytic process, owing to the non-equilibrium nature of cold plasma towards easily activate molecules [24,25]. The cold plasma, driven by renewable electricity, could generate abundant energetic electrons during the discharge process [26,27]. Next that, the energetic electrons collide with inert chemical bonds in molecules to form highly relative species such as ions and radicals under nearly ambient conditions, which further induce the occurrence of heterogeneous reactions, especially thermodynamically limited reactions [28,29].

Recent studies on plasma-catalytic polyolefin recycling have predominantly focused on the formation of gas fuels and aromatics via a two-stage reactor system [30,31]. Huang et al. developed a plasma-catalytic tandem reactor, where polyethylene pyrolysis coupled with plasma-enhanced HZSM-5 catalysis, achieved 51.32 % liquid yield and 98.43 % aromatic selectivity through parameter optimization [32]. Chen et al. demonstrated that plasma-catalytic PP pyrolysis over the ZSM-5 catalyst increased the yield of gaseous products to 47 %, concurrently improved benzene, toluene and xylene (BTX) formation and suppressed coke deposition compared to conventional pyrolysis methods [33]. However, this two-stage reactor suffered from low energy utilization due to the input of external heat by electric furnace. Shi et al. designed the single stage reactor configuration without external heating and realized complete LDPE conversion with 62.5 % oil yield and 100 % aromatic selectivity through CO_2-enhanced plasma catalysis using Ga/Hie-ZSM-5 catalyst [34]. Whereas, the separated packing mode of Ga/Hie-ZSM-5 and LDPE could make the short-lived reactive species quench before reaching catalyst surface, which was not beneficial to the generated intermediates converted into the target product timely. Therefore, developing highly efficient plasma-catalytic systems for polyolefin upcycling has become critically imperative.

Herein, the Sm/USY catalysts were prepared via a deposition-precipitation method, and mixed with LDPE to achieve the plastic waste fully conversion in an innovatively plasma-assisted catalytic system without external heating. The addition of Sm_2O_3 species into USY support could modulate the Lewis/Brønsted acid ratio for the catalysts, which further affect the product distribution and selectivity. For the optimized 5Sm/USY catalyst, the highest liquid yield with a carbon number range of C_4-C_{15} was obtained, up to 83 % at an input power of ca. 56 W. Meanwhile, the energy cost of the applied plasma-catalytic LDPE upcycling mode was greatly lower than that of the traditionally thermal-catalytic LDPE upcycling mode. The developed plasma-catalytic system demonstrated broad applicability to other polyolefin wastes and real plastic wastes, especially yielding 88 % recoverable liquid hydrocarbons from real plastics. In addition, the reaction mechanism for plasma-catalytic LDPE upcycling over the 5Sm/USY catalyst was proposed. This study identifies a feasible route for plasma-assisted polyolefin wastes upcycling to produce valuable oil fuels with high yield under relatively mild condition.

2. Experimental section

2.1. Catalyst preparation and materials

2.1.1. Catalyst preparation

Sm/USY catalysts were prepared by deposition-precipitation method. In general, 1.5 g of USY zeolite (SiO_2/Al_2O_3=40) was dissolved in 100 mL of deionized water at room temperature, followed by adding 221.7 mg of $Sm(NO_3)_3$ and stirring for 30 min. Then, a certain amount of NaOH solution was added dropwise until the pH of the above mixed solution was 9.25 and stirred vigorously for 4 h. Afterwards, the mixed solution was filtrated and the obtained precursor was placed in a vacuum drying oven at 60 °C for 12 h to evaporate the aqueous solvent. After that, the 5Sm/USY catalysts was obtained by calcining the dried solids at 550 °C for 6 h with a temperature rate of 5 °C/min. The prepared samples are donated as xSm/USY, where x is the mass percentage of Sm (x = 1, 3, 5, 7). At the same time, the Sm/USY catalysts prepared by other deposition-precipitation methods were only replaced with Na_2CO_3, NH_4HCO_3 as precipitants, and other steps remained unchanged.

In comparison, the 5Sm/USY catalyst was prepared by the incipient wetness (IWI) method. Typically, 221.7 mg of $Sm(NO_3)_3$ was dissolved in deionized water and fully impregnated on USY zeolite, and then placed for 48 h. Subsequently, the impregnated sample was dried at 60 °C for 12 h and calcined at 550 °C for 6 h. The obtained catalyst is donated as Sm/USY-IWI. Moreover, the same mass fraction of 5 M/USY (M=Cr, Mn, Fe, Co, Ni, Cu, Zn, La, Ce, Pr, Y, Gd; 5 wt% loading) was prepared by the same preparation method.

Additionally, Sm/USY was prepared by ion exchange (IE) method. $Sm(NO_3)_3$ and USY zeolite were added together in 100 mL of deionized water and stirred in a water bath at 40 °C for 24 h. Subsequently, Sm/USY-IE was obtained by filtration, drying and calcination.

Meanwhile, the Sm_2O_3 catalyst was synthesized via calcination of a $Sm(OH)_3$ precursor. Specifically, NaOH solution was dropwise added to $Sm(NO_3)_3$ solution and stirred for 4 h. The resulting $Sm(OH)_3$ precipitate was collected via filtration, then dried in a vacuum at 80°C for 12 h. The precursor was subsequently calcined at 550°C for 6 h to obtain the final Sm_2O_3 catalyst. And, the Sm_2O_3+USY catalyst was prepared by a mechanical mixing method. Briefly, 65 mg Sm_2O_3 sample was deeply physically mixed with 1.425 g USY powder in an agate mortar for 60 min to obtain the the Sm_2O_3+USY catalyst. In the preparation of NaOH-USY catalyst, NaOH solution was firstly added dropwise to the USY solution until the pH was 9.25. The mixed solution was stirred for 4 h, then NaOH-USY catalyst was obtained by filtration, drying and calcination.

2.1.2. Polyolefins samples

The polyolefin samples were provided by Guangyuan Plasticizer Co., China. Furthermore, LDPE exhibits molecular masses of 4×10^3 g/mol and 1×10^5 g/mol, while HDPE and PP have molecular masses of 1.2×10^5 g/mol and 1×10^4 g/mol, respectively. Throughout this work, except for experiments aimed at revealing the universality of the 5Sm/USY catalyst in plasma catalytic hydrogenolysis of recycled waste plastics, all other experiments used the LDPE with a molecular weight of 4×10^3 g/mol to evaluate catalytic performance.

2.2. Catalyst characterization

The X-ray diffraction (XRD) patterns were obtained using an XRD-6000 diffractometer (Shimadzu, Kyoto, Japan) with Cu Kα radiation (λ = 0.1524 nm) to confirm crystalline phases for the catalysts. The system set to 30 kV and 30 mA was taken over the 2θ in the range of 5–70°.The information of crystalline phases was obtained by comparing XRD patterns to standard cards.

Nitrogen adsorption-desorption was measured at −196 °C on Micromeritics ASAP 2460 analyzer. The Brunauer-Emmett-Teller (BET) was used to calculate the total surface area. The pore volume and pore size of them were also measured on the same instrument. Prior to the measurements, the samples should be degassed at 200 °C for 4 h under vacuum condition.

The actual Sm content of catalysts was analyzed using inductively coupled plasma optical emission spectrometry (ICP-OES) on ICP-Optical Emission spectrometer (Model 2000, PerkinElmer).

Transmission Electron Microscope (TEM) images of the catalysts were recorded on a Tecnai G2 F30 S-Twin system (Thermo Scientific, America) operated at 300 kV. Prior to the measurement, the catalyst was dispersed in absolute ethanol for 30 min under ultra-sonication condition and subsequently dripped on a copper grid.

Thermogravimetric measurements (TG-DTG) of spent catalysts were performed on a TGA 2 thermobalance (Mettler Toledo). Prior to the test, 8 mg spent catalyst was pretreated by Ar stream at 200 °C for 30 min and then cooled down to 40 °C. After that, the temperature was increased from 40 to 800 °C with a ramping rate of 10 °C/min in 10 % O_2/Ar flow to measure the weight loss of the spent catalyst.

Ammonia temperature-programmed desorption (NH_3-TPD) measurements were performed using an infrared absorption spectrometer (SICK-MAIHAK-S710; Germany). Prior to analysis, the catalyst samples underwent pretreatment at 300 °C for 1 h under Ar flow (100 mL/min), followed by cooling to 50 °C. The samples were then exposed to a 10 % NH_3/Ar mixture until adsorption saturation was achieved, with subsequent Ar purging for 2 h. Temperature-programmed desorption was conducted by heating the samples from 50 °C to 800 °C at a controlled heating rate of 10 °C/min under continuous Ar flow (100 mL/min). The outlet NH_3 concentration was continuously monitored in real-time during the thermal desorption process.

Pyridine-infrared (Py-IR) spectroscopic analysis was conducted on a Bruker Tensor 27 FTIR spectrometer equipped with a vacuum reaction cell. Prior to measurement, the catalyst samples underwent sequential pretreatment: N_2 purge at 200 °C for 30 min under dynamic vacuum, followed by in-situ pyridine vapor adsorption at 150 °C for 30 min. Spectral acquisition was performed at 150 °C with 60 scan accumulations at 4 cm^{-1} resolutions. The acid site distribution was quantified using the integrated molar extinction coefficients of characteristic vibrational bands (1545 cm^{-1} for Brønsted acid, 1450 cm^{-1} for Lewis acid) through the following relationships [35]:

$$C_B = 1.88\, A_B R^2 / W \quad (1)$$

$$C_L = 1.42\, A_L R^2 / W \quad (2)$$

where C_B and C_L are the acid concentrations per catalyst mass (mmol/g), A_B and A_L are the integrated peak areas of Brønsted acid sites and Lewis acid sites (μmol/cm), respectively; R is the radius of the catalyst disk (cm), and W is the mass of the catalyst disk (mg).

X-ray photoelectron spectroscopy (XPS) characterization was conducted on a spectrometer (ESCALAB 250XI, Thermo VG, America) with Al K_α source (1486.6 eV) operated at 15 kV and 10 mA. The binding energy values of Sm species were corrected using the C 1 *s* peak centered at 284.8 eV.

^1H and ^{13}C nuclear magnetic resonance (NMR) spectra were recorded on a Bruker AVANCE NEO 400 MHz and 600 MHz spectrometers, respectively. $CDCl_3$ containing 0.03 % (v/v) tetramethylsilane (TMS) was employed as the locking solvent. All measurements were conducted at 298 K with a 5 mm PABBO BB-1H/D Z-GRD probe. Chemical shifts (δ) were referenced to the residual solvent peak ($CDCl_3$: δH 7.26 ppm, δC 77.16 ppm) with TMS as the internal standard (δ = 0 ppm).

Oxygen temperature-programmed oxidation (O_2-TPO) analysis was conducted using an infrared absorption spectrometer (SICK-MAIHAK S810, Germany). The spent 5Sm/USY catalyst (0.025 g) was heated from 30 to 800 °C in a 5 vol% O_2/ N_2 flow at a linear heating rate of 10 °C·min^{-1}. CO_2 and CO concentrations were monitored online during the oxidation process.

2.3. Catalyst activity evaluation

2.3.1. Experimental setup

The experiments were carried out in a dielectric barrier discharge (DBD) reactor for a single-stage plasma-catalytic polyolefins upgrading (Fig. 1). The reactor was composed of gas controller, a plasma generator, a plasma reactor, a cooling system, gas sampling bags and analysis detectors. The reactor configuration was constructed as a cylindrical corundum ceramic tube (8 mm inner diameter, 2 mm wall thickness) functioning as the dielectric layer. A grounded stainless-steel mesh encircling the external surface served as the outer electrode, while a 3 mm diameter stainless-steel rod centered along the longitudinal axis functioned as the inner electrode, linked to a high-voltage AC power supply. Within this configuration, a discharge gap of 2.5 mm was maintained with a 10 cm plasma discharge region. Additionally, the corundum tube was encased in a thermal insulation layer, and a K-type thermocouple was placed on the outer wall of the reactor tube to continuously monitor the temperature rise in the discharge region. A high-voltage probe attached to the inner electrode was used to measure the applied voltage. Meanwhile, the outer electrode circuit incorporated an external capacitor (C_{ext} = 3.3 nF) for quantitative assessment of charge transfer during the discharge process. Electrical signals from both electrodes were simultaneously acquired via a digital oscilloscope (Tektronix TBS 2000 Series) operating at 100 MHz sampling frequency. The discharge power was calculated from Lissajous figures using Q-V plot integration methodology.

2.3.2. Plasma-catalytic activity evaluation

As depicted in Fig. 1, the plasma-catalytic upgrading process for polyolefin upgradation was conducted in a DBD reactor. Before activity test, the catalyst was pretreated at 300 °C for 2 h under pure Ar flow (30 mL/min). Then, 500 mg LDPE particles (40–60 mesh) and 200 mg

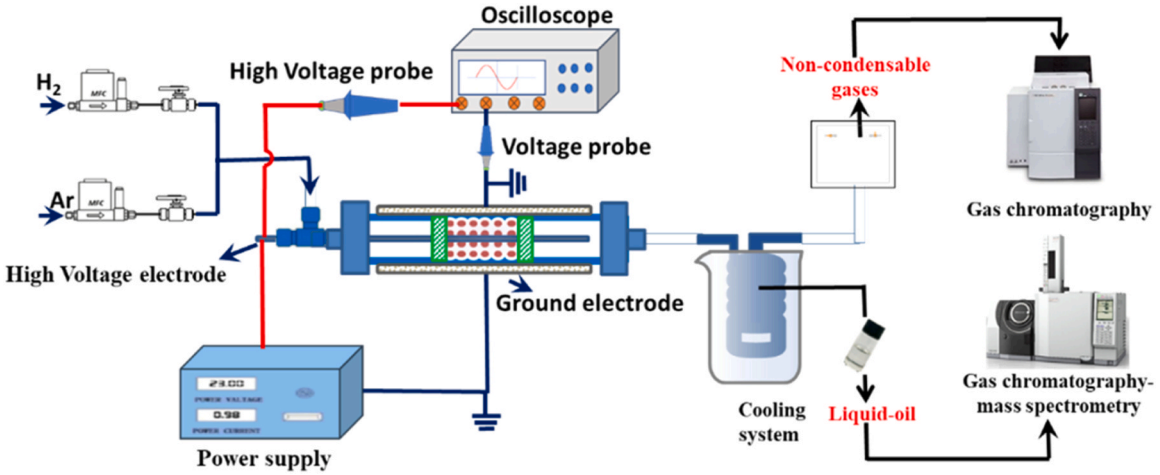


Fig. 1. Schematic diagram for plasma-catalytic upcycling of polyethylene waste reaction.

(40–60 mesh) pretreated catalyst were fully mixed and packed in the plasma zone. The reactor was continuously purged with Ar (50 mL/min) under ambient temperature conditions for 30 min to achieve oxygen-free environment. The catalytic upgrading process was maintained with 30 % H_2/Ar feed gas (30 mL/min) at an input power of ca.56 W for 1 h without external heating. During the reaction process, a liquid cooled condenser incorporated into the exhaust line continuously collected condensable products, while gaseous samples were charged in Tedlar gas sampling bags. A blank experiment with no packing catalysts was also performed and all reaction parameters were consistent.

2.3.3. Thermal-catalytic activity evaluation

The thermal-catalytic activity of the applied catalysts was evaluated in a pyrolysis reactor for polyolefins upgradation. Prior to the experiment, the catalyst was pretreated by Ar (30 mL/min) 2 h thermal pretreatment at 300 °C for 2 h. 500 mg LDPE sample (40–60 mesh) and 200 mg pretreated catalyst (40–60 mesh) were fully mixed and uniformly distributed within the thermal zone. An Ar flow of 50 mL/min was introduced into the tube at room temperature for 30 min to remove the air in tube. Subsequently, the thermal-catalytic activity test was conducted with a total flow rate of 30 mL/min (30 % H_2/Ar) at 550 °C for 1 h. During the reaction process, a condenser and a gas sampling bag were attached to collect the liquid and gaseous products, respectively.

2.4. Methods of product analysis

The gaseous products collected in gas sampling bags underwent comprehensive composition analysis on a Shimadzu GC-2014 system equipped: a flame ionization detector (FID) and a PLOT-Al_2O_3 column to quantify hydrocarbon products, with methane serving as the internal standard. To differentiate between endogenous and exogenous methane through depolymerization pathways, samples were subjected to comparative chromatographic analysis before and after standard injection. The calibration of FID was performed using standard mixtures with varying volume fractions by the external standard method.

Liquid products were analyzed on a SHIMADZU GMS-QO2020NX chromatograph equipped with an AOC-20i Plus. The chromatography was equipped with an SH-Rxi-5Sil MS phase column (SHIMADZU, 30 m×0.25 mm, film thickness 0.25 μm) using He as carrier gas. The mass-to-charge ratio (m/z) of each component in the liquid products was analyzed using mass spectrometry within the range of 30–500 m/z. The identification of peaks was achieved by comparing their retention times to those of standard aromatic compounds, with reference to the mass spectral library (NIST 11S). Due to the complexity of the components in the products, the relative content of the liquid products was determined using peak area normalization. The mass of the liquid product was obtained by weighing the difference in mass before and after the condenser tube.

Overall, the total product was divided into gas, liquid, and wax remaining in the reaction tube. The weight of gaseous products (m_{gas}) was calculated by the total amount of them determined via GC analysis. The weight of the liquid products (m_{liquid}) was calculated by the difference between condenser before reaction and condenser after reaction. The weight of wax products (m_{wax}) was calculated by subtracting the mass of liquids and gases in the product from the mass of waste plastic input ($m_{plastic}$). The yield of various products from the catalytic degradation of polyethylene was determined using the following equation:

$$m_{wax} = m_{plastic} - m_{liquid} - m_{gas} \tag{3}$$

$$Y_{gas}(wt\%) = \frac{m_{gas}}{m_{LDPE}} \times 100\% \tag{4}$$

$$Y_{liquid}(wt\%) = \frac{m_{liquid}}{m_{LDPE}} \times 100\% \tag{5}$$

$$Y_{wax}(wt\%) = 100\% - Y_{gas} - Y_{liquid} \tag{6}$$

Discharge power ($P_{discharge}$) was calculated as follow:

$$P_{discharge} = f \int_0^T u dQ \tag{7}$$

Where f refers to the frequency of discharge; Q refers to the charge of discharge; u refers to the applied voltage.

The relative selectivity of each component (S_i) in liquid products were presented in the following equation:

$$S_i = \frac{A_i}{\sum A_i} \times 100\%\# \tag{8}$$

Where A_i refers to the peak area of corresponding liquid component.

The degree of branching was defined as the fraction of isomerized hydrocarbons (C_{iso}) among all hydrocarbon products (C_n and C_{iso}) [36]:

$$\text{Degree of branching}(\%) = \frac{C_{iso}}{C_{iso} + C_n} \times 100\% \tag{9}$$

Where C_{iso} refers to iso-alkanes and iso-alkenes, while C_n refers n-alkanes and n-alkenes.

The average rate of C-C bond scission ($R_{C\text{-}C}$) was determined using following equation [37]:

$$R_{C-C} = \frac{N_t - N_0}{t} \tag{10}$$

Where N_0 and N_t represent the total number of chains at time 0 and time t, respectively.

Energy consumption (E) was calculated by the following formula:

$$E = \frac{N_1 - N_0}{m_{plastic}} \tag{11}$$

Where N_1 and N_0 refer to the final reading and initial reading of the electricity meter, respectively; $m_{plastic}$ refers to the mass of waste plastic.

3. Result and discussion

3.1. Characterization of the catalysts

The Sm/USY catalysts (SiO_2/Al_2O_3=40) with different Sm loadings (1 wt%, 3 wt%, 5 wt% and 7 wt%) were prepared by deposition-precipitation method. The XRD patterns of the catalysts were shown in Fig. 2(A). It was noted that the characteristic diffraction peaks at the range from 10 to 35° were observed for all the samples, ascribing to the crystal planes of USY zeolites with cubic structure (JCPDS NO. 01–077–1150) [38]. The observed distinct peaks with relatively strong intensity suggested the good crystallinity of the prepared Sm/USY catalysts and the USY zeolite structure was preserved after adding Sm_2O_3 species. However, the intensity of diffraction peaks corresponding to pristine USY zeolite was reduced with the increase of Sm_2O_3 loadings, which the slight change of the zeolite framework crystallinity induced by Sm_2O_3 incorporation. The characteristic diffraction peaks attributed to Sm_2O_3 phase were not observed even in the 7Sm/USY catalyst with highest Sm loading, indicating a high dispersion of Sm_2O_3 species presented in all prepared samples.

The preservation of the textural properties of the 1Sm/USY, 3Sm/USY, 5Sm/USY, and 7Sm/USY catalysts were further confirmed by N_2 adsorption-desorption isotherms and pore size distributions. As shown in Fig. 2(B), all the isotherms exhibited characteristic type IV with H4 hysteresis loop at the P/P_0 ranged from 0.45 to 0.99, suggesting the coexistence of the mesoporous and microporous channels for all the samples [39]. Analyzed by the pore size distribution (Fig. S1), the catalysts comprised both mesopores and macropores, with the mesopores predominantly concentrated within a pore diameter range of 2.5–4.5 nm. The Sm-loaded catalysts exhibited distinct textural modifications compared to pristine USY zeolite, summarized in Table 1. Both the specific surface areas and the pore volumes of the prepared catalysts

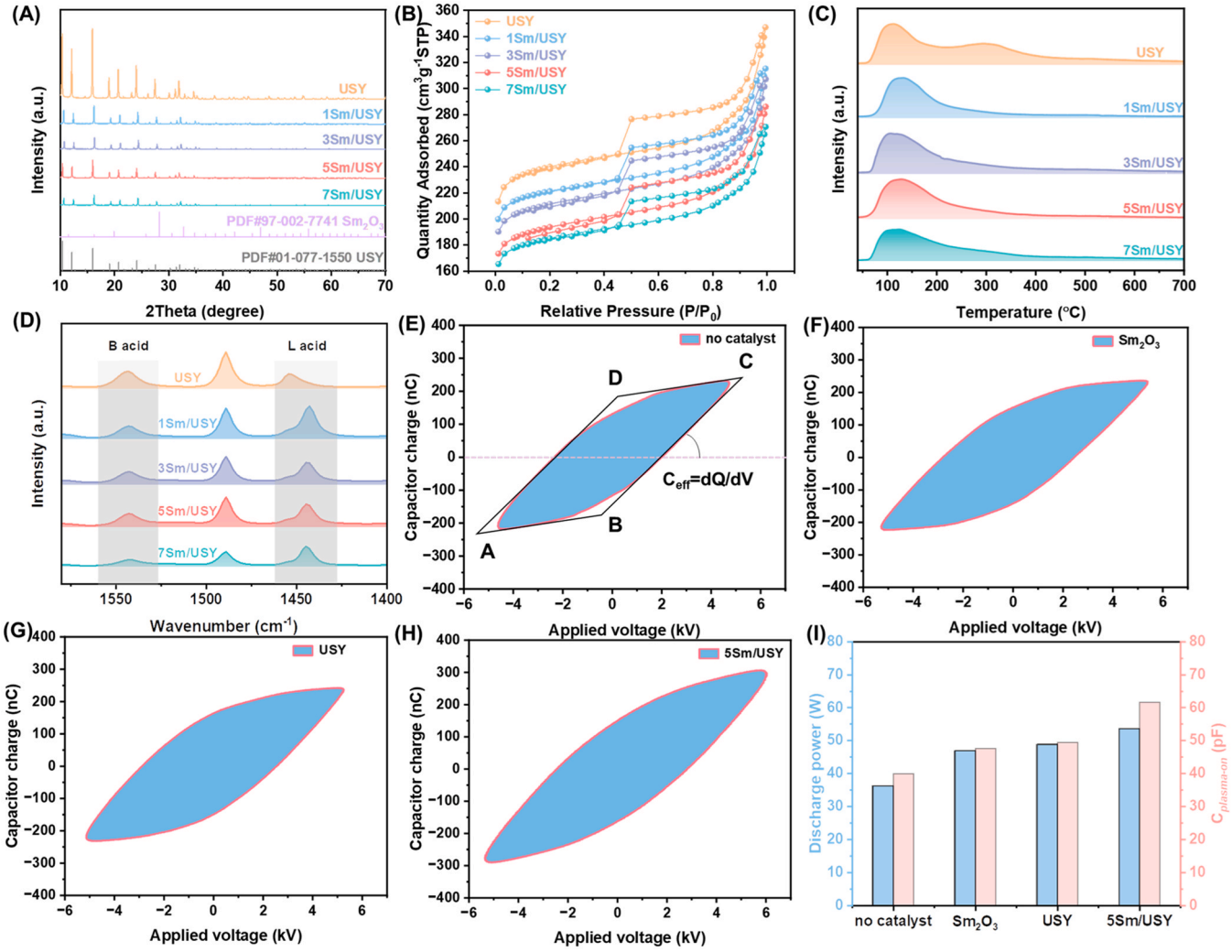


Fig. 2. (A) XRD patterns, (B) N_2 adsorption-desorption isotherms, (C) NH_3-TPD profiles and (D) Py-IR spectra measured at 150 °C of Sm/USY catalysts with different Sm loadings. Q-U Lissajous figures of (E) no catalyst, (F) Sm_2O_3 catalyst, (G) USY catalyst and (H) 5Sm/USY catalyst in plasma-catalytic reaction. (I) Discharge power and effective capacitance of no catalyst, Sm_2O_3 catalyst, USY catalyst and 5Sm/USY catalyst in plasma-catalytic reaction. (Reaction conditions: 0.5 g LDPE, 0.2 g catalyst, input power of ca. 56 W, 30 mL/min, H_2: Ar=3:7, 1 h, without external heating, ambient pressure.).

Table 1
Textural properties and compositions of USY, 1Sm/USY, 3Sm/USY, 5Sm/USY and 7Sm/USY catalysts.

Sample	S_{BET} (m^2/g)[a]	V_{Total} (cm^3/g)[a]	V_{Micro} (cm^3/g)[a]	V_{Meso} (cm^3/g)[a]	Sm loading (wt%)[b]
USY	716.1	0.54	0.31	0.23	0
1Sm/USY	671.4	0.48	0.29	0.19	0.9
3Sm/USY	642.8	0.48	0.27	0.21	2.5
5Sm/USY	639.5	0.48	0.27	0.21	4.2
7Sm/USY	563.5	0.41	0.23	0.18	5.8

[a] Determined by N_2 adsorption-desorption isotherms.
[b] The content of Sm was measured by ICP-OES result.

decreased in the following order: USY> 1Sm/USY> 3Sm/USY> 5Sm/USY> 7Sm/USY, suggesting the incorporation of Sm_2O_3 species into USY zeolite greatly resulted in the decrease of both the specific surface areas and the pore volumes of them. This trend highlighted the critical threshold of Sm content for maintaining structural integrity in modified zeolite catalysts. Meanwhile, the actual Sm loading over the 5Sm/USY catalyst was 4.2 wt% in Table 1.

Further, the NH_3-TPD profiles revealed fundamental insights into the acid properties of Sm-modified USY zeolites (Fig. 2(C)). All samples mainly showed characteristically bimodal desorption peaks, with the low-temperature peak (60–250 °C) and the high-temperature feature (250–500 °C). As shown in Table 2, the addition of Sm_2O_3 species into USY resulted in a decrease of total acid amounts compared to the pure USY zeolite, calculated by the total ammonia desorption peak area, which may be due to the fact that some of the acidic sites in USY were covered by Sm_2O_3 species. Meanwhile, the attenuation of high-temperature peak intensity with increasing Sm loading suggested that Sm_2O_3, noted as a moderately basic oxide, inhibited NH_3 adsorption. In

Table 2
Textural properties of USY, 1Sm/USY, 3Sm/USY, 5Sm/USY and 7Sm/USY catalysts.

Sample	Acidity	Acidity(mmol/g)[b]		
	(μmol NH_3/g·Cat)[a]	Brønsted	Lewis	L/B
USY	654	0.094	0.045	0.479
1Sm/USY	423	0.046	0.090	1.957
3Sm/USY	534	0.048	0.078	1.625
5Sm/USY	512	0.059	0.075	1.271
7Sm/USY	479	0.029	0.067	2.310

[a] Determined by NH_3-TPD.
[b] Determined by Py-IR at 150 °C.

addition, Py-IR analysis (Fig. 2(D)) quantified the changes of Brønsted (B) acid sites and Lewis (L) acid sites after Sm modification. With the increase of Sm loading for the xSm/USY catalysts, the amounts of B acid sites decreased substantially from 0.094 mmol/g to 0.029 mmol/g and the corresponding L acid sites increased from 0.045 mmol/g to 0.067 mmol/g, indicating the induction of Sm_2O_3 species on the USY support predominantly generated L acid sites. Due to the difference in Si-O-Al polarizability induced by Sm_2O_3 species, the amount B acid sites and the L acid sites over Sm/USY catalysts varied, which in turn affected the L/B ratio. Remarkably, 7Sm/USY showed 67 % B acid sites loss versus only 29 % L acid sites gain, indicating Sm_2O_3 clusters physically block B acid sites beyond chemical modification effect. Similar result has also been reported by Wang et al. [40], it was found that the increase in L acid sites with increasing Y loading into dealuminated BEA zeolite, concurrently accompanied by the decrease in OH nest-bounded pyridine, which revealed Y species could perturb the silanol nest structure, resulting in a loss of them.

The Lissajous figures over different packing materials were analyzed to investigate the discharge properties during the plasma process, shown in Fig. 2(E-H). At an input power of ca. 56 W, the Lissajous graph presented a parallelogram shape of pure plasma (without parking any inert materials) and packing Sm_2O_3 (Fig. 2(E, F)), indicating the discharge mode was filamentary discharge. Differently, when the catalysts were packed into the plasma region, especially for the 5Sm/USY catalyst, the shape of the Lissajous graph changed to oval-like shape, indicating that the discharge mode was transformed from filamentary discharge to surface discharge, further manifested the catalytic role of the catalyst [41]. And, the DBD effective capacitance (slope of the BC and AD lines in Lissajous graph) and the discharge power were compared, shown in Fig. 2(I). It was observed that the discharge power and corresponding effective capacitance over different materials decreased as the following order: 5Sm/USY> USY> Sm_2O_3> no catalyst, indicating the addition of catalyst increased the discharge power. In particular, the discharge power was further increased in the case of the 5Sm/USY catalyst, suggesting the addition of Sm_2O_3 changed the discharge property of the catalyst, namely promoted the charge transport and accumulation, thereby obtained the synergistic enhancement between plasma and catalyst. As reported by Li et al. that rare earths species had excellent electron transport, which further accelerated charge transferring density and helps catalyze the reaction [42].

Furthermore, the high-resolution TEM images of the 5Sm/USY catalyst (Fig. 3(A-C)) showed the preserved porous architecture of USY support and presented no obvious sintering of Sm_2O_3, which was in good agreement with N_2 sorption analysis and XRD results (Fig. 2(A, B)). In addition, the elemental mappings exhibited the homogeneous dispersion of Sm_2O_3 species on the support. This was also a good indication of the strong interaction between Sm_2O_3 species and USY zeolite. The chemical state of the 5Sm/USY catalyst was investigated using XPS, shown in Fig. S2. Only Sm $3d_{5/2}$ peak and Sm $3d_{3/2}$ peak corresponding to the Sm^{3+} species centered at 1086.3 eV and 1113.8 eV [43], respectively, were clearly observed, indicating the formation of Sm_2O_3. The $3d_{5/2}$ and $3d_{5/2}$ peaks of Sm in the Sm_2O_3 catalyst were located at 1086.7 eV and 1114.3 eV, respectively. Compared to the Sm_2O_3 catalyst, the binding energy of the Sm $3d_{5/2}$ and $3d_{5/2}$ spectra over the USY supported Sm_2O_3 catalyst negatively shifted to 0.4 and 0.5 eV, respectively. It indicated the Sm-O bonds in Sm_2O_3 became weaker for the 5Sm/USY catalyst, which was ascribed to the high dispersion of Sm_2O_3 on the USY support, deriving from the strong interaction between Sm_2O_3 and USY.

3.2. *Effect of Sm loading for plasma-catalytic LDPE hydrogenolysis*

Firstly, thermogravimetric (TG) analysis fundamentally revealed the feasibility of USY zeolite catalyst for PE cracking. Pure LDPE displayed on onset degradation temperature of 350 °C and was fully degraded at 490 °C (Fig. S3). However, both the corresponding onset temperature (200 °C) and full degradation temperature (400 °C) were greatly decreased after adding the USY zeolite catalyst to the process (the catalyst/LDPE mass ratio of 2/5), indicating the effective promotion of LDPE cracking by USY zeolite catalyst. Subsequently, the reaction conditions for plasma-catalytic LDPE degradation were systematically investigated including input power, carrier gas, and packing mode of catalyst and plastic. As shown in Fig. S4, the optimized reaction parameters were an input power of 56 W, 30 % H_2/Ar and one stage

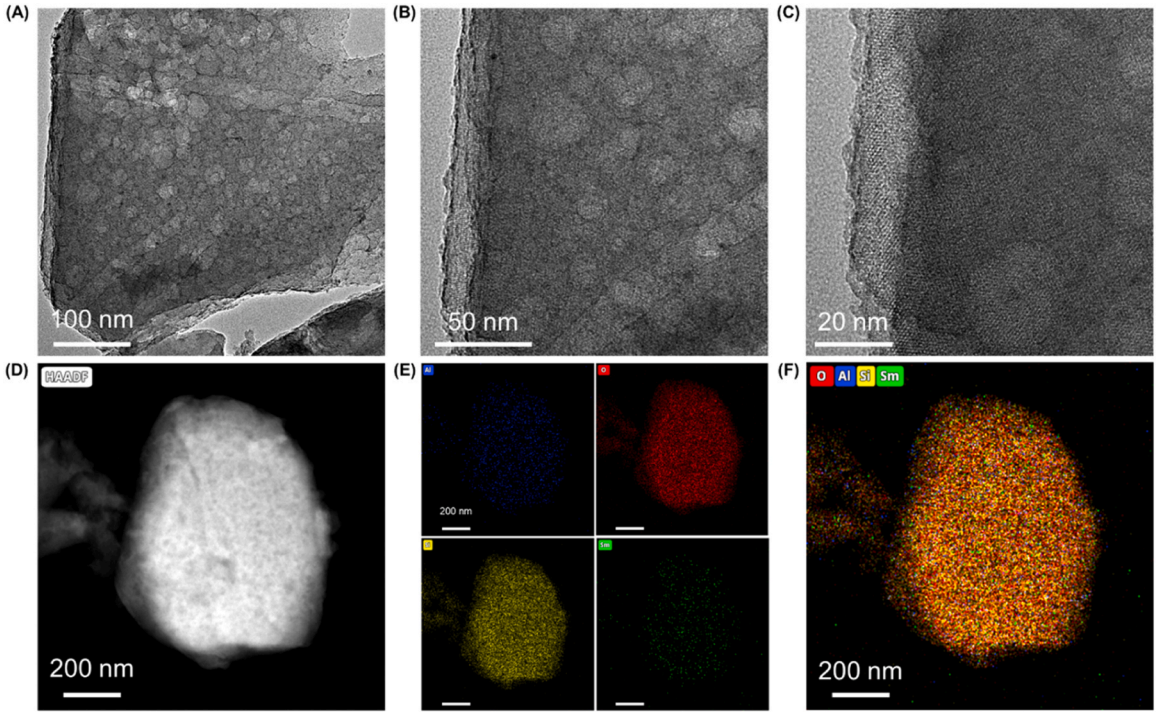


Fig. 3. (A-C) TEM images and (D-F) the corresponding mapping of 5Sm/USY catalysts.

packing mode, respectively, owing to obtained the maximum yield of liquid product. And, the plasma-catalytic activity of the catalysts was evaluated under above optimized reaction conditions. Furthermore, the active metal types for the 5 M/USY catalysts (M=Cr, Mn, Fe, Co, Ni, Cu, Zn, La, Ce, Pr, Y, Gd) were also optimized for plasma-assisted reaction (Fig. S5). Compared to the USY zeolite, the 5Mn/USY and 5Ni/USY catalysts exhibited higher yield of wax product (Fig. S5(A)), suggesting the negative effect of Mn and Ni species toward the catalytic performance. In contrast, the modification of Cr, Fe, Co, Cu, Zn, La, Ce, Pr, Sm, Y, and Gd species decreased the wax yield. Especially, adding Sm into USY support (5Sm/USY catalyst) achieved a dramatic wax reduction from 93 mg to 5 mg. And, the yield of liquid product over these catalysts were also compared (Fig. S5(B)), using USY zeolite as the standard. It was observed that the 5Zn/USY catalyst exhibited a comparable liquid yield to metal-free USY catalyst, while the 5Sm/USY presented the highest yield of liquid product (59 %) with the mass increased from 273 mg to 295 mg. The above results revealed that the 5Sm/USY catalyst had a better catalytic activity in plasma-assisted LDPE recycling reaction, which was ascribed to achieve reduce the formation of wax byproducts and improve liquid hydrocarbons recovery.

Next that, the condensation bath for the liquid product collection was optimized by replacing the ice-water bath with a liquid nitrogen-isopropanol bath, which maintained a low-temperature condition. The yield of liquid products over the 5Sm/USY catalyst collected in liquid nitrogen-isopropanol bath obviously increased to 80 % (402 mg) compared to that in ice-water bath (Fig. S6(A)). Furthermore, as shown in Fig. S6(B), the evaluation results of 5Sm/USY catalysts, synthesized via incipient wetness impregnation, ion exchange, Na_2CO_3-, NaOH-, and NH_4HCO_3-assisted deposition-precipitation methods, respectively, revealed that the catalyst prepared by the NaOH-assisted deposition precipitation method had the highest liquid yield, up to 83 % liquid yield with LDPE complete transformation. Based on the experimental results, the optimized condensation way was a liquid nitrogen-isopropanol bath, and the optimized preparation method of applied catalysts was the NaOH-assisted deposition precipitation method. Besides, lower flow rate (10 mL/min) reduced mass transfer rates and occurred more coke deposition (Fig. S7(A)), while higher flow rate (50 mL/min) prevented adequate contact with the catalyst, all that led to poor catalytic performance in Fig. S7(B). However, the mobility of polymeric chains and adequate contact between the LDPE and the catalyst were obtained at a moderate flow rate (30 mL/min), further improving the catalytic performance and achieving the highest liquid

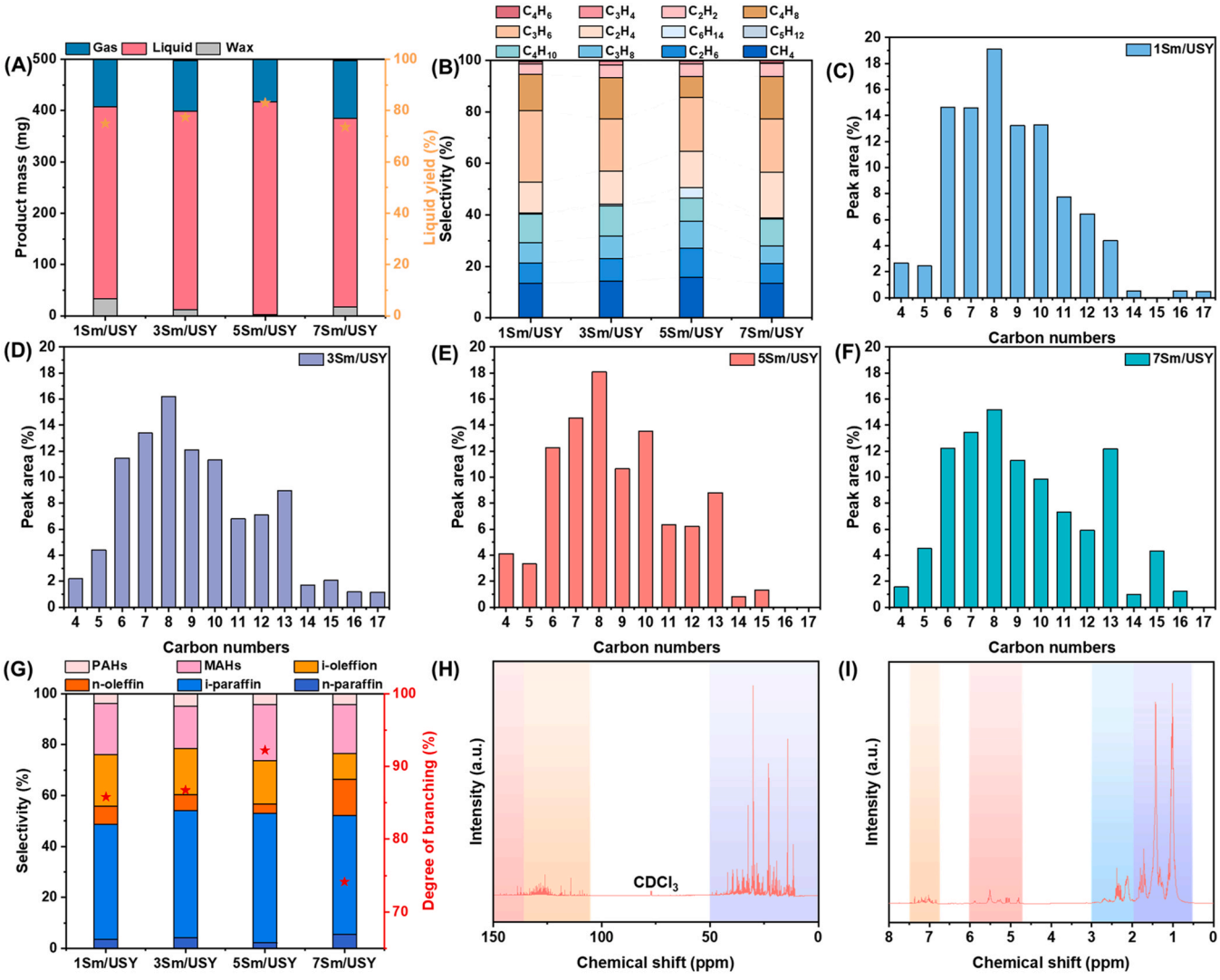


Fig. 4. Plasma-catalytic performance of Sm/USY catalysts with different Sm loadings: (A) product mass, (B) the selectivity of different components in gaseous product, (C-F) carbon number distribution in liquid product, (G) the selectivity of different components in liquid product. (H) ^{13}C NMR and (I) ^{1}H NMR spectra of liquid products over 5Sm/USY catalyst for plasma-catalytic LDPE hydrogenolysis. (Reaction conditions: 0.5 g LDPE, 0.2 g catalyst, input power of ca. 56 W, 30 mL/min, H_2:Ar=3:7, 1 h, without external heating, ambient pressure.).

yield. And as shown in Fig. S8 (A-B), from the viewpoint of efficient plastic upcycling, the optimized mass ratio of 5Sm/USY catalyst to LDPE was 2:5 compared for plasma catalytic hydrogenolysis reaction. During comparing with different plastic particle size, the highest liquid yield over the 5Sm/USY catalyst was obtained during the upcycling of LDPE with the particle size of 40–60 mesh. Based on these experimental results, the optimal reaction conditions were a gas flow rate of 30 mL/min, a catalyst-to-plastic ratio of 2:5, and plastic particle sizes of 40–60 mesh, with the best plasma-catalytic upgrading of recycled plastic over the 5Sm/USY catalyst.

Under these optimized conditions, the effects of Sm loading towards catalytic performance over Sm/USY catalysts were investigated for plasma-assisted LDPE hydrogenolysis (Fig. 4). The actual temperature of the reactor outer wall was measured, shown in Fig. S9. It was noted that the temperature rise was about 550 °C at an input power of ca. 56 W for 60 min, which exhibited the advantage of a quasi-adiabatic plasma-catalytic reactor. For the Sm/USY catalysts, the amount of wax product initially reduced and then increased slowly thereafter as the loading of Sm_2O_3 species increased, whereas the formation of liquid product had an inverse trend (Fig. 4(A)). Among them, the 5Sm/USY catalyst showed the highest catalytic performance with liquid production of 415 mg, significantly suppressing the generation of both wax and gaseous products. This highlighted the dual efficiency of 5Sm/USY catalyst in enhancing targeted liquid product yield and minimizing undesirable side reactions. Note that, the total alkanes selectivity in the gaseous products versus Sm loading exhibited a volcanic curve (Fig. 4(B)), and the highest selectivity of total gaseous alkanes was observed over the 5Sm/USY catalyst, suggesting enhanced hydrogenation efficiency at medium Sm loading. In addition, as illustrated in Fig. 4(C-F), the carbon number distribution of liquid products over the Sm/USY catalysts were similar, mainly varying from C_4 to C_{13} centered at C_8, although the Sm loadings were different. These results demonstrated that the carbon number distribution strongly dependent on the shape-selective effect of parent USY support rather than Sm_2O_3 species. The alkane fractions in liquid product were dominated over all Sm/USY catalysts (Fig. 4(G)). 1Sm/USY, 3Sm/USY and 5Sm/USY catalysts had high isomerization degree, especially the 5Sm/USY catalyst with the highest isomerization degree of 92 %, but 7Sm/USY catalyst blocked more acid sites, leading to a decreased in isomerization.

Furthermore, ^{13}C NMR and ^{1}H NMR characterizations of liquid product over the 5Sm/USY catalyst were employed to analyze the hydrocarbon's structure. The ^{13}C NMR spectrum revealed that aliphatic carbon species (0–50 ppm) constituted the primary components, accompanied by trace amounts of protonated aromatic/olefinic carbons (95–136 ppm) and non-protonated aromatic/olefinic carbons (136–160 ppm) (Fig. 4(F)) [44,45]. Meanwhile, the ^{1}H NMR result exhibited characteristic peaks ascribed to aliphatic protons at a chemical shift of 0.5–1.85 ppm, accompanied with weak peaks corresponding to olefinic/aromatic protons (1.85–2.8 ppm), olefinic protons (4.6–6.0 ppm), and aromatic protons (6.7–8.0 ppm) (Fig. 4(G)) [46,47]. These results further confirmed that the aliphatic hydrocarbons in liquid product were predominant. In addition, the octane number of liquid product in plasma-catalytic LDPE upcycling process was nearly same compared to commercial gasoline (86.6 versus 89.5 %) (Table S1).

In order to analyze the structure change and coke of the spent catalysts, XRD and TG characterizations were carried out. As evidenced by the XRD patterns in Fig. S10(A), the spent Sm/USY catalysts with different Sm loadings retained their characteristic crystallinity, confirming the preservation of structural integrity over them during the reaction process. Meanwhile, compared to other spent catalysts, the spent 5Sm/USY catalyst exhibited the lowest weight loss of 2.89 wt%, indicating the addition of Sm_2O_3 species with the optimized loading (5 wt%) could suppress the formation of coke deposits (Fig. S10(B)), which further enhanced the catalytic performance.

3.3. The synergy of plasma catalysis for LDPE hydrogenolysis

The LDPE hydrogenolysis experiments over the optimized 5Sm/USY catalyst were performed under three reaction modes, namely pure plasma, thermal catalysis and plasma catalysis, to investigated the role of plasma without external heating (Fig. 5). For the reaction under pure plasma condition, the yield of liquid and wax products was 50 % and 26 % (Fig. 5(A)), respectively, indicating the pure plasma could dissociate the chemical bonds in LDPE to generate molecules with low unit chains to some extent via the electrons effect of plasma. Differently, the addition of 5Sm/USY catalyst under plasma-catalytic condition greatly improved the yield of liquid product (83 %) and simultaneously no wax product was generated, suggesting the catalytic role of the catalyst brought into play and further promote the LDPE conversion to targeted liquid hydrocarbons, which also exhibited the synergy between the catalyst and plasma under relatively mild conditions. In contrast, the thermal-catalytic LDPE hydrogenolysis over the 5Sm/USY catalyst produced lower liquid yield (73 %) accompanied with higher amount of wax (17 %) at 550 ℃, which further revealed the advantage of plasma catalysis in HDPE upcycling. In addition, it was noted that the selectivity of olefin components in liquid product under pure plasma condition was 78 % (Fig. 5(B)), which further indicated that the plasma was more inclined to break the C-H bond in LDPE. However, the i-alkanes in liquid product over the 5Sm/USY catalyst under both plasma-catalytic and thermal-catalytic conditions were dominant, which were ascribed to the promotion of isomerization and β-scission processes on acid sites of the catalyst. Especially, the plasma-catalytic mode also showed the highest degree of branching (92 %).

For the gaseous product, higher CH_4 selectivity under both pure plasma and plasma-catalytic conditions was observed compared to that under thermal-catalytic condition (Fig. 5(C)), indicating plasma was preferential cleavage of terminal C-C bonds in LDPE. As shown in Fig. 5 (D), the carbon number distribution in liquid product under pure plasma condition ranged from C_4 to C_{19}. In contrast, employing the 5Sm/USY catalyst with its distinct shape-selective nature showed a narrower carbon number distribution for both thermal catalysis and plasma catalysis. Especially, in plasma-catalytic mode, the carbon number distribution varying from C_4 to C_{15}, with a 98 % selectivity of gasoline component (C_4-C_{13}). Notably, the plasma-catalytic hybrid system achieved balanced performance: maintaining high liquid yield and i-alkanes selectivity through synergistic plasma-generated reactive species and zeolite-mediated acid catalysis, demonstrating successful integration of physical excitation by electrons and chemical activation pathways by catalysts.

In addition, TG-DTG tests of the spent 5Sm/USY catalysts were conducted to evaluate the coke amount under different reaction modes. As shown in Fig. 5(E), the carbon species could be different according to the oxidation temperature ranging from 300 to 700 °C [48,49]: (i) low-temperature oxidation (<500 °C) corresponding to amorphous carbonaceous deposits, (ii) intermediate decomposition (500–650 °C) indicating filamentous carbon nanostructures, and (iii) high-temperature mass loss (>650 °C) representing graphitic carbon species (Fig. 5(E)). It was observed that the carbon species deposited on the spent 5Sm/USY catalyst under thermal-catalytic condition contained amorphous and filamentous carbon species, while that under plasma-catalytic condition was mainly filamentous carbon species (Fig. 5(F)), which verified the stronger C-C cleavage ability in plasma. However, the spent catalyst after plasma-assisted reaction has a great weight loss of 2.89 wt% compared to that after thermal-catalytic reaction (6.54 wt%). This result further suggested that the selectively activate chemical bonds in LDPE by coupling plasma and the specific catalyst facilitated to inhibit the formation of carbon species during the reaction process.

The C-C bond scission rates under three reaction modes were also calculated, shown in Fig. 5(G). Compared to the thermal catalysis, pure plasma-assisted LDPE cycling had a stronger bond-breaking capacity

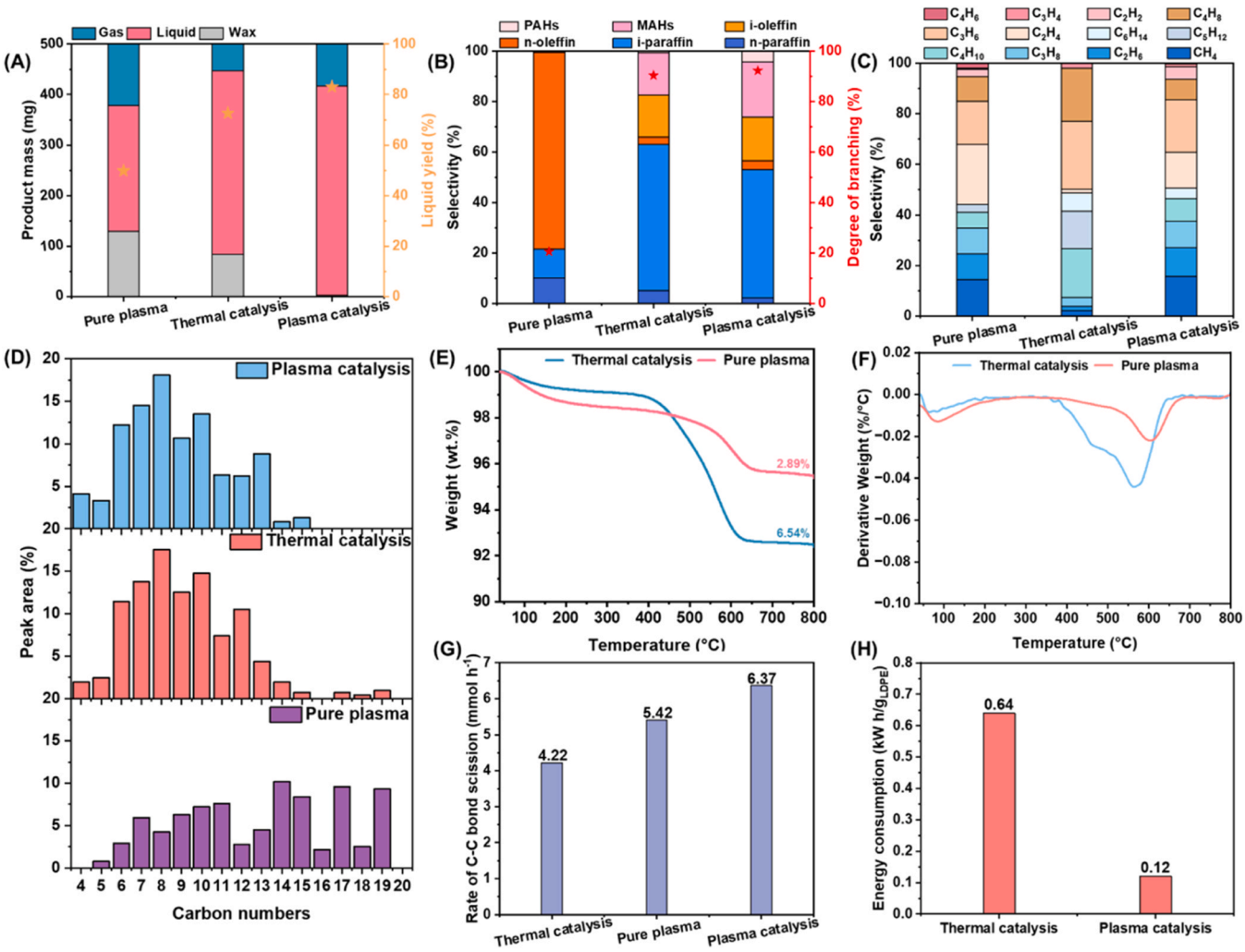


Fig. 5. Pure plasma performance, thermal catalysis and plasma catalysis performance over 5Sm/USY catalyst: (A) product weight, (B) the selectivity of different components in liquid product, (C) the selectivity of different components in gaseous product, (D) carbon number distribution in liquid product. (E) TG and (F) DTG curves of spent 5Sm/USY catalysts, (G) rate of C-C bond scission of plasma, thermal catalysis and plasma catalysis and (H) electricity consumption of thermal catalysis and plasma catalysis over 5Sm/USY catalyst. (Reaction conditions of plasma: 0.5 g LDPE, 0.2 g catalyst or not, input power of ca. 56 W, 30 mL/min, H_2: Ar=3:7, 1 h, without external heating, ambient pressure. Reaction conditions of pyrolysis: 0.5 g LDPE, 0.2 g catalyst, 30 mL/min, 550°C, 1 h.).

due to higher C-C bond cleavage rates apparently. Further, the plasma-catalytic system achieved a maximum cleavage rate of 6.37 mmol/h, demonstrating plasma coupled with catalyst enabled performance optimization and significantly reduced the formation of wax product with longer chain units, giving fully play to the roles of physical excitation in plasma and chemical activation in catalyst. More importantly, the compared results of energy consumption under thermal-catalytic and plasma-catalytic modes showed that the plasma catalysis system demonstrated greatly lower energy consumption in LDPE conversion, achieving 0.12 kW·h/g_{LDPE} (Fig. 5(H)). In contrast, energy consumption reached 0.64 kW·h/g_{LDPE} in conventionally thermal catalysis, which was 5.3 times that of plasma catalysis. This substantial disparity in energy consumption not only quantified the advantage of energy efficiency in plasma-catalytic mode but also underscored its potential as a sustainable alternative to conventional energy-intensive approaches.

3.4. *The proximity of Sm_2O_3 and USY for plasma-catalytic LDPE hydrogenolysis*

The catalytic performance of Sm_2O_3, USY, mechanically mixed Sm_2O_3+USY catalysts were evaluated to investigate the effect of proximity degree between Sm_2O_3 species and USY zeolite for plasma-catalytic HDPE hydrogenolysis. Firstly, XRD patterns of the applied catalysts showed that the mechanically mixed Sm_2O_3+USY catalyst had a similar characteristic diffraction peak (2θ of 10–35°) compared to the USY catalyst (Fig. S11(A)), confirming structural preservation of the zeolite framework after Sm_2O_3 mixing. Significantly, no distinct XRD peaks attributable to crystalline Sm_2O_3 phase (JCPDS No. 97–002–7741) was observed, which indicated the high dispersion of Sm_2O_3 on the USY surface. In contrast, reference Sm_2O_3 sample exhibited characteristic diffraction peaks at 2θ= 28.25° and 32.89°, corresponding to Sm_2O_3 phase (JCPDS No. 97–002–7741). Next that, NH_3-TPD and Py-IR were also performed to determine the acidity property of the catalysts. As shown in Fig. S11(B), the Sm_2O_3+USY catalyst and USY catalyst exhibited the similar NH_3 desorption peaks, suggesting the mixing of Sm_2O_3 on USY was not change the acid strength of the zeolite support, which also verified the Sm_2O_3 dispersed on the USY surface rather than entered in the USY pores. In contrast, the acidity strength of the 5Sm/USY catalyst became weaker due to the NH_3 desorption peak shifted to lower temperature, revealing the Sm_2O_3 species entered into the pore channels of USY and thereby affected the acid sites, achieving a nanoscale proximity between Sm_2O_3 species and USY zeolite. Meanwhile, compared to the USY zeolite, the content of L acid sites increased after doping with Sm_2O_3 species (Fig. S11(C), which

resulted from the L acid formation induced by Sm_2O_3 species. More importantly, the amount of L acid sites over the 5Sm/USY catalyst prepared by deposition precipitation method were significantly higher than that over the Sm_2O_3+USY catalyst (Table S2). These results conclusively demonstrated strong interactions between Sm_2O_3 and USY zeolite.

The catalytic performance of the above catalysts was also evaluated under plasma-assisted condition, shown in Fig. 6. Compared to pure plasma without packing catalysts, the Sm_2O_3 and USY zeolite exhibited higher yield of liquid product (Fig. 6(A)). And, the mechanically mixed Sm_2O_3+USY catalyst slightly improved the yield of liquid product. Differently, the highest liquid yield was obtained over the 5Sm/USY catalyst. This observation suggested that a close nanoscale proximity of Sm_2O_3 species and USY zeolite promoted the formation of liquid hydrocarbons. Meanwhile, Figs. 6(B) and 6(C) showed distinct product distributions in gaseous and liquid products, respectively. USY-containing catalysts showed higher alkane concentrations, particularly evident in liquid products where iso-alkanes dominated and degree of branching were higher above 87 %, demonstrating B acid site-mediated isomerization. The highest yield of n-alkanes in the liquid product over the Sm_2O_3 catalyst in comparison to that over other catalysts, suggesting the hydrogenation process could be enhanced by Sm_2O_3 catalysis.

From the carbon number distribution of liquid product over the different catalysts (Fig. 6(D-F)), it was clearly observed that the carbon

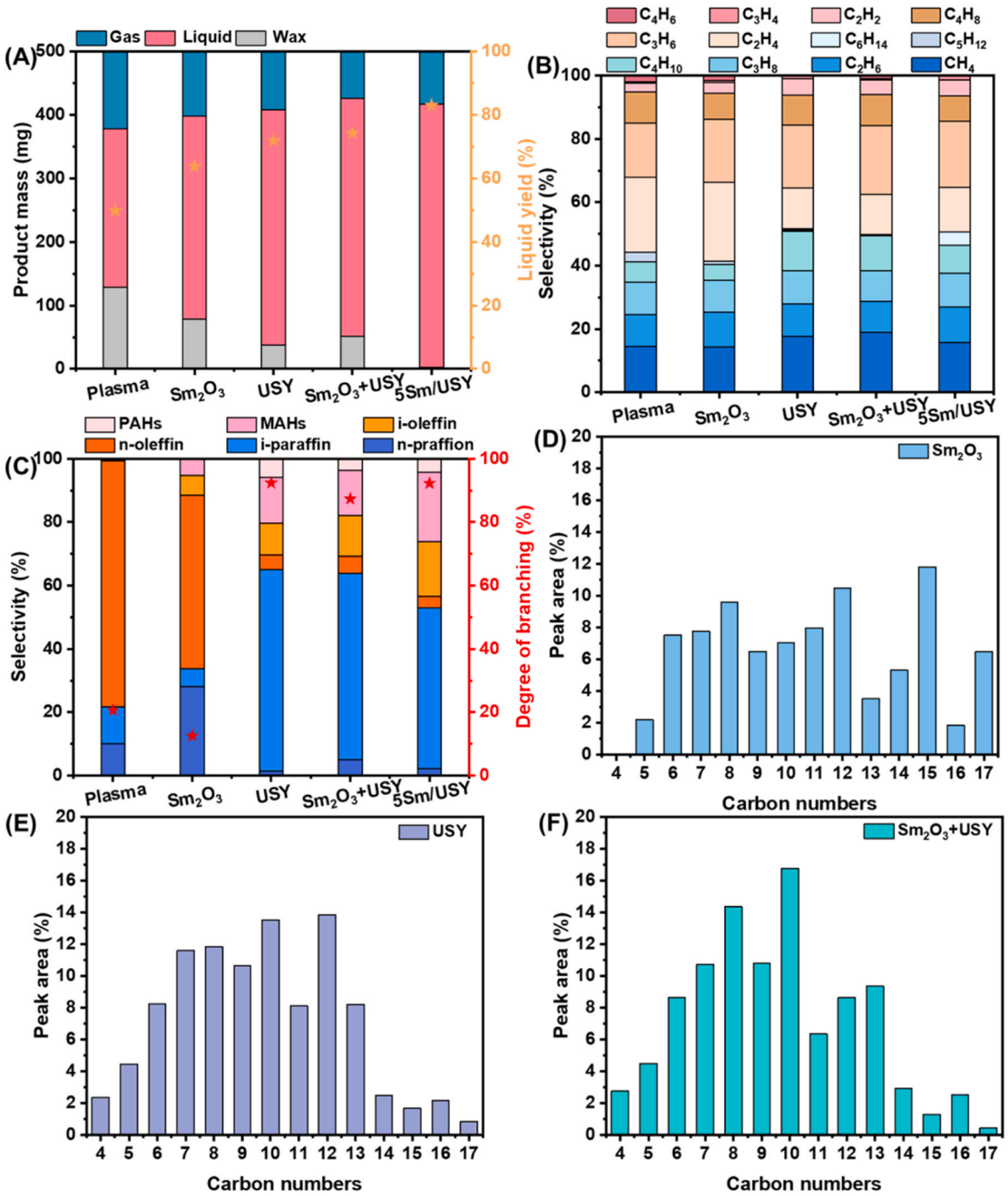


Fig. 6. Plasma-catalytic performance of Sm_2O_3, USY, Sm_2O_3+USY and 5Sm/USY catalysts: (A) product mass, (B) the selectivity of different components in gaseous product, (C) the selectivity of different components in liquid product and (D-F) carbon number distribution in liquid product. (Reaction conditions: 0.5 g LDPE, 0.2 g catalyst, input power of ca. 56 W, 30 mL/min, H_2:Ar=3:7, 1 h, without external heating, ambient pressure.).

number over the Sm_2O_3 catalyst ranged from C_5 to C_{17}, which was similar to that under plasma-only condition (C_5-C_{19}), indicating that the acid sites on Sm_2O_3 could cleave the C-C bonds in LDPE but limited extent, with wax still present. However, the narrower carbon number distributions over USY-containing catalysts were achieved, ascribing to the shape-selective effect of their porous architecture, which imposed molecular size constraints. Furthermore, compared to USY zeolite catalyst, the Sm_2O_3+USY catalyst showed a lower carbon number centered at C_{10} than the parent USY catalyst (centered at C_{12}), originated from the ability of Sm_2O_3 species towards C-C bonds cleavage. Especially, the 5Sm/USY catalyst exhibited the most pronounced carbon number reduction centered at C_8 (Fig, 4(C)), confirming the plasma-catalytic synergy and the close proximity of acid sites improved the LDPE cracking efficiency. These results proved the importance of the proximity of Sm_2O_3 and USY. Additionally, XRD and TG characterizations of the spent catalysts were performed, shown in Fig. S12. The distinct peaks with relatively strong intensity were also observed of all the spent catalysts, suggesting the catalyst structures seemed to be not changed during the reaction process (Fig. S12(A)). It was observed from the TG (Fig. S12(B)) result that the Sm_2O_3 catalyst experienced a significant mass loss before 400 °C, which was attributed to the oxidation of produced wax on the catalyst. However, only a 0.7 wt% mass loss was present at the oxidation temperature of above 400 °C, demonstrating the Sm_2O_3 catalyst inhibited the formation of coke during the reaction process. Furthermore, compared to the USY catalyst with more B acid sites (7.50 wt%), mechanically mixed Sm_2O_3+USY and 5Sm/USY catalysts reduced the amount of carbon deposition, which further reflected the coke-resistant ability of Sm_2O_3 species in the LDPE hydrogenolysis.

3.5. *Proposed reaction pathway for plasma-catalytic hydrogenolysis of polyethylene*

In the catalytic upcycling of polyolefins, it is generally accepted that the acid sites on catalysts are the primary active sites for the chemical bonds cleavage. Wherein, the L acid sites could activate C-H bonds in polyolefin macromolecules adsorbed on catalysts, accompanied with in situ generate H species, and B acid sites exhibit excellent C-C bonds cleavage ability, which synergistically promote the polyolefin conversion process [18,50–53]. Meanwhile, increasing the number of the L acid sites could enhance the polyolefins adsorption and improve the activate rate of C-H bonds, also facilitating to enhance the B acid strength neighboring hydroxyl sites via interacting with ligand oxygen [54]. According to the experimental results, the Sm species affected the amount and strength of acid sites (Table 2). As shown in Fig. 7(A), the L/B ratios over xSm/USY catalysts were correlated with the liquid yield, obtaining a volcano-shaped curve. Note that, the USY catalyst exhibited the lowest liquid yield attributed to the weak activation of C-H bonds on insufficient L acid sites. In contrast, the introduction of Sm_2O_3 species increased the number of L acid sites and thereby improved the activation of C-H bonds. Moreover, as the Sm loading increasing from 1 wt% to 5 wt%, the amount of B acid site increased and L acid site content decreased of the xSm/USY catalysts, with a subsequent increase in the ability to crack C-C bonds and a slight change of L/B ratio. The highest liquid yield was obtained over 5Sm/USY catalyst under plasma condition, and the corresponding L/B ratio was 1.271 with the stronger synergistic effect between L acid sites and B acid sites.

In order to investigate the effect of NaOH solution towards both catalyst structure and catalytic activity during the preparation of Sm/

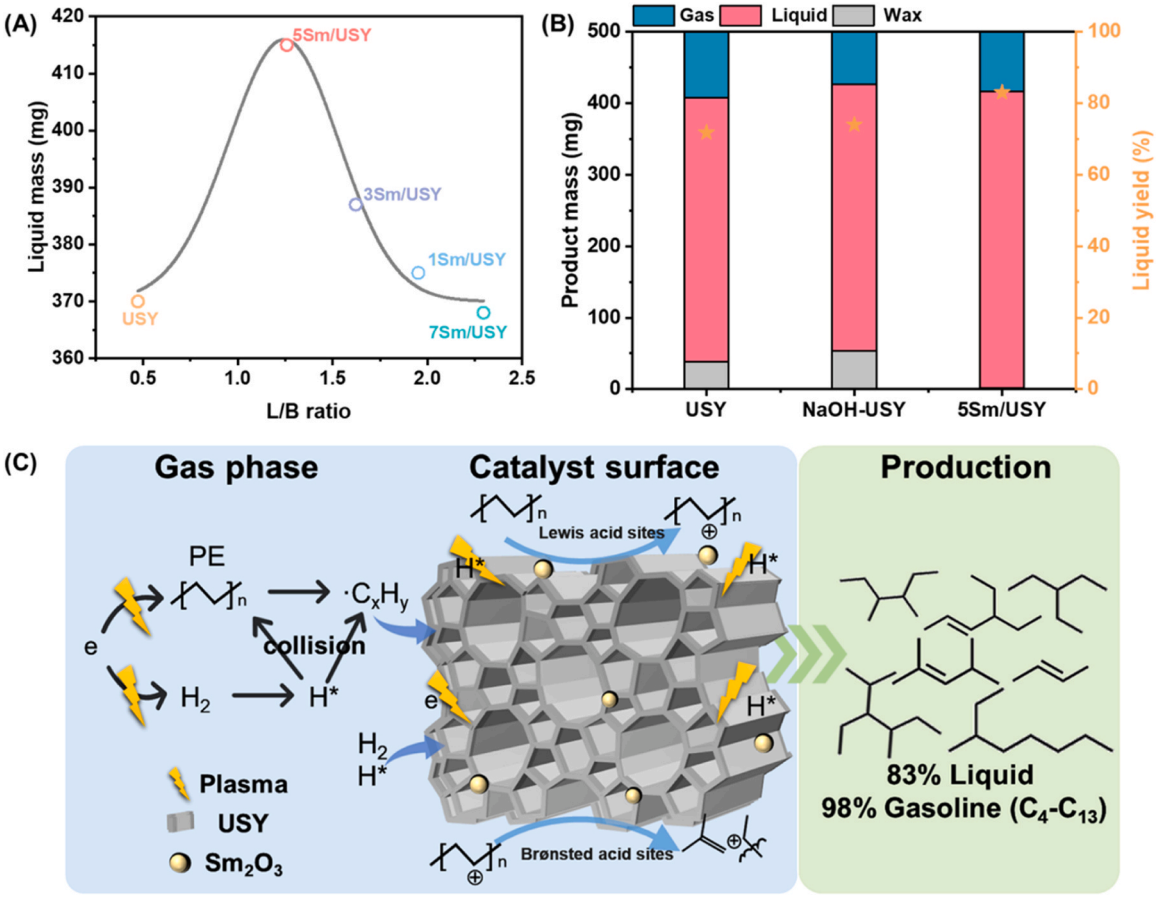


Fig. 7. (A) Liquid mass over L/B ratio of xSm/USY catalysts and (B) Product mass of plasma-catalytic performance over USY, NaOH-USY and 5Sm/USY catalysts. (Reaction conditions: 0.5 g plastics, 0.2 g catalyst, input power of ca. 56 W, 30 mL/min, H_2:Ar=3:7, 1 h.) (C) A possible mechanism of plasma-catalytic LDPE upcycling over the 5Sm/USY catalyst without external heating.

USY catalyst by deposition precipitation method, the same method was used to prepare NaOH-USY catalyst without adding Sm_2O_3 species. Compared to the USY catalyst, NaOH-USY catalyst had the sample characteristic diffraction peak, indicating that it maintained the framework of the USY catalyst after the pretreatment of NaOH solution (Fig. S13(A)). However, the intensity of the diffraction peaks at 10–35° was decreased by comparison to the original USY catalyst, which was due to the fact that alkaline treatment reduced crystallinity of zeolite [55]. Besides, NH_3-TPD and Py-IR measurements were also conducted to determine the acidic properties of the NaOH-USY catalyst. As shown in Fig. S13(B), the peak intensity corresponding to the medium NH_3 desorption over the NaOH-USY catalyst greatly reduced than that over parent USY catalyst, which was similar to the result that desilication of NaOH solution reduced zeolitic acidity reported by Datka et al. [56]. Meanwhile, the NaOH-USY catalysts had lower B and L acid content than USY catalyst (Table S2). Quantitatively, the amount of total acid sites decreased in the following order: USY> 5Sm/USY> NaOH-USY, further suggesting the addition of Sm_2O_3 species provided the acid sites.

In comparison, NaOH-USY catalyst was evaluated for plasma-catalytic LDPE hydrogenolysis. The liquid yield of the NaOH-USY catalyst (372 mg) was slightly higher than USY catalyst (370 mg) (Fig. 7(B)), which was consisted with the trend of L/B ratio associated with liquid yield. However, higher amount of waxes was produced over the NaOH-USY catalyst, which was ascribed to its lower acid concentration. The product distribution of gas and liquid over NaOH-USY catalyst was similar to that over USY catalyst (Fig. S14-15). In contrast, the 5Sm/USY catalyst, which sacrificed a little of B acid sites but greatly increased L acid sites in comparison to the NaOH-USY catalyst, showed a great increase of liquid yield, again illustrating the importance of L acid site in abstraction of a hydride ion to activate C-H bonds. Similar result has been reported by Han et al. [18,52] that L acid site had high activity to activate C-H bonds of PE.

Compared to the USY catalyst, the addition of Sm_2O_3 species increased L acid sites, but the distance between Sm and USY was also particularly important. Wang et al. [57] also showed that the distance between Ru nanoclusters and FeO_x nanoclusters was crucial by comparing the activity of catalysts $FeRu/CeO_2$ sample prepared by impregnation with physically mixed Ru/CeO_2 and Fe/CeO_2 sample, and found that the catalyst prepared by physical mixed method showed lower liquid yield. Similarly, compared to Sm_2O_3+USY catalyst, the 5Sm/USY catalyst could rapidly transfer the dehydrogenated species from the L acid sites to the B acid sites or molecular sieve pores during the reaction process, which accelerated the reaction and promoted the formation of liquid hydrocarbons.

Based on the above results, a reaction mechanism for the plasma-catalytic LDPE hydrogenolysis over the 5Sm/USY catalyst was proposed (Fig. 7(C)). Firstly, the heat generated by the plasma during the adiabatic reaction transferred to plastic and catalyst, with rapidly melting and transforming into a solid-liquid contact state, which contributed the depolymerization of the plastic and catalytic process. According to the study purported by Li et al., H_2 plasma could provide highly reactive hydrogen radicals and hydrogen atoms that promoted the reaction under mild conditions [58]. Yao et al. found that during PS hydrogenolysis in pure plasma condition, the reactive H* species produced by plasma-activated H_2 hydrogenated with the·C_xH_y species produced by PS cleavage to form liquid product. Meanwhile, the electrons collided with C_xH_y species to generate smaller C_xH_y radicals [59]. Based on the above analysis, for this plasma-catalytic LDPE hydrogenolysis process, H* species was formed by plasma-induced H_2 activation in the gas phase, simultaneously the generated electrons by plasma discharge activated and dissociated the C-C and C-H bonds in LDPE by means of inelastic collisions, also resulting in producing hydrocarbon fragments that include shortened polymer chains, C_xH_y radicals, and H* species. The activity of the catalysts also affected the performance of the catalyst [60]. As reported by Lisak et al. [61], carbocations can be generated by the protonation of hydrocarbons on B acid sites or the abstraction of a hydride ion by L acid sites. The L acid site synergized with the B acid sites under plasma catalytic conditions. LDPE were also activated abstraction of a hydride ion to activate C-H bonds and generated carbocations on L acid site in the catalyst surface reaction process. Then, carbocations occurred skeletal isomerization and β-scissions to produce shorter carbocations and i-oleffins on B acid site. Finally, i-paraffins with higher selectivity were produced through hydrogenated with formed H_2 or H* species. Furthermore, hydrocarbons with a carbon number ranging from C_4 to C_{15} were produced due to the shape-selective effects of pore structures in the catalyst.

3.6. Stability and versatility of the 5Sm/USY catalyst

The stability of the optimized 5Sm/USY catalyst was evaluated by successive reaction cycles (Fig. 8). The liquid yield remained above 80 % during the first three cycles although it slightly decreased in the fourth cycle (Fig. 8(A)), which demonstrated the relatively good recyclability. The selectivity of the alkanes in both the gas and liquid products decreased with the increase of cycle number, and the isomerization of the liquid products also decreased, from 92 % in the first to 74 % in the fourth reaction (Fig. 8(B-C)), which might be ascribed to coke deposited on the catalyst to blocking B acid sites and impairing hydrogenation functionality. Nevertheless, the carbon number distribution of liquid products predominantly remained within C_4-C_{13} (Fig. 8(D-E)). XRD characterization of the recycled catalysts confirmed retention of the USY zeolite framework (Fig. S12(A)), although crystallinity decreased compared to the first cycle, attributed to accumulated carbon deposits. TG/DTG analysis revealed the formation of coke in the fourth cycle, especially the weight loss of the spent catalyst after four cycles was 9.59 wt%, verifying formed carbon species covered the B acid sites and further inhibited isomerization process (Fig. S16(B-C)). Further, as shown in Fig. S17, the 5Sm/USY catalyst after four cycles contained amorphous carbon deposits (<500 °C), fibrous carbon nanostructures (500–650 °C) and a small amount of graphitic carbon species (>650 °C) [47,48], which blocked the pores and covered the active sites to decrease catalytic activity, consistent with the results of S16 (C).

Evaluation of LDPE (Mw = 4000 g/mol) upcycling demonstrated the high activity and selectivity of the 5Sm/USY catalyst under plasma-assisted conditions without external heating, achieving 83 % liquid yield with 98 % gasoline. Further experiments were performed for plasma-catalytic upcycling of different types of plastics to reveal the versatility of the 5Sm/USY catalyst (Fig. 9), including polyolefins (LDPE-10 w, Mw = 100,000 g/mol; HDPE, Mw = 120,000 g/mol; PP, Mw = 10,000 g/mol) and real-world plastics. The various plastics were completely converted and the yield of liquid product ranged from 83 % to 90 % without external heating (Fig. 9(A)). The result might also be an advantage under plasma conditions, where the rapid activation of C-H bonds and C-C bonds cracking by energetic electrons generated by plasma discharge. Meanwhile, all waste plastics exhibited similar compositional distributions in gaseous products (Fig. 9(B)), containing substantial methane, ethylene, and propylene, which may be attributed to the preferential cleavage of C-H and terminal C-C bonds under plasma conditions. As demonstrated in Fig. 9(C), liquid products derived from plastic upcycling showed isomerization degrees exceeding 90 %, with PP degradation exhibiting the highest isomerization efficiency (95 %). It likely originated from the branched structure of PP, which facilitated the formation of tertiary carbocation intermediates and enhances bond cleavage tendency during isomerization [62]. Furthermore, the 5Sm/USY catalyst exhibited exceptional performance for real-world plastics waste conversion, achieving 88 % liquid yield with 95 % isomerization selectivity in the products. The carbon number distribution of liquid products was predominantly concentrated in the C_4-C_{13} range in Fig. S18. These fully demonstrated the excellent activity and universality of the 5Sm/USY catalyst in plasma-catalytic waste plastics upcycling.

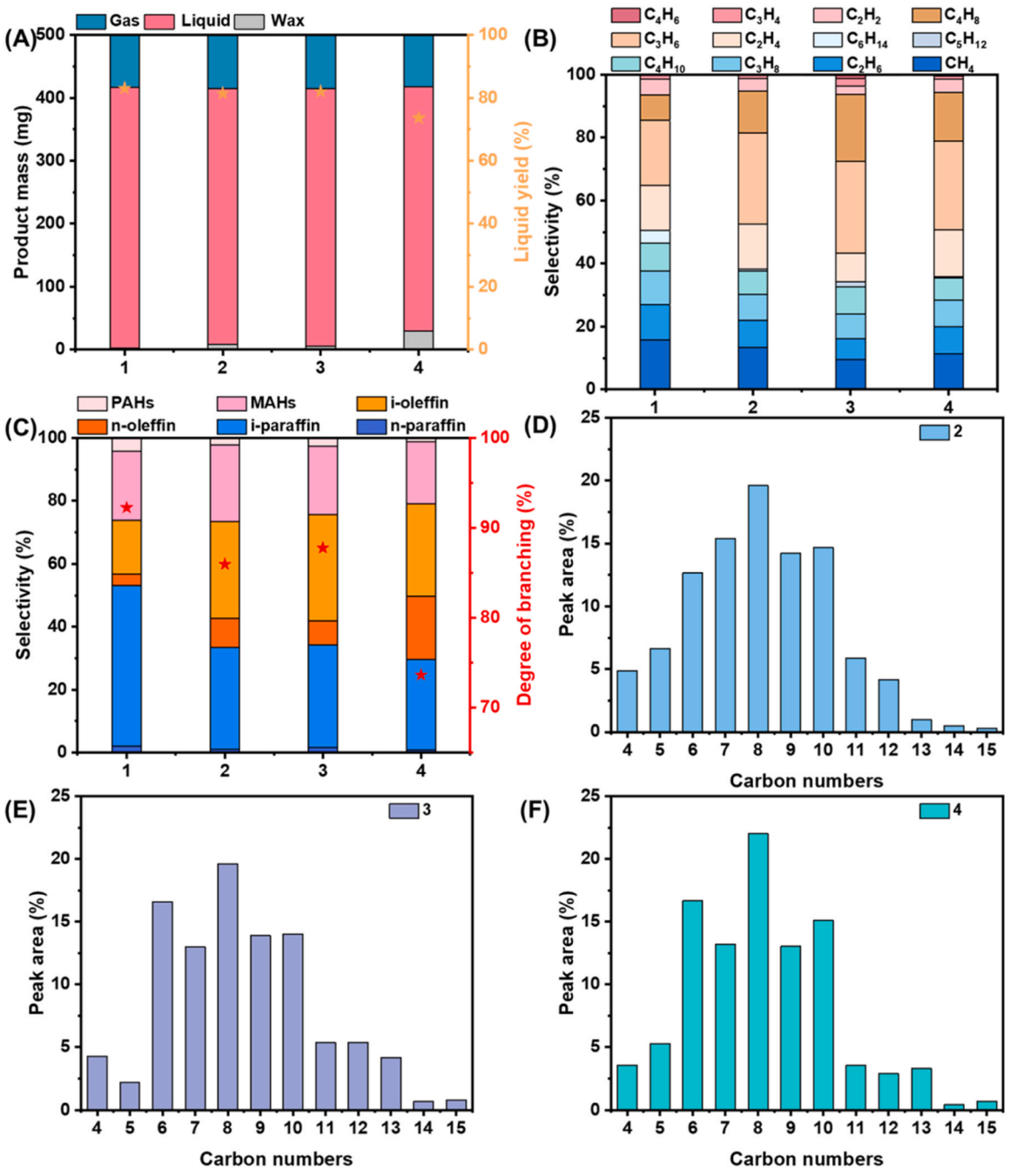


Fig. 8. Plasma-catalytic performance of successive cycles over 5Sm/USY catalyst: (A) product mass, (B) the selectivity of different components in gaseous product, (C) the selectivity of different components in liquid product and (D-F) carbon number distribution in liquid product. (Reaction conditions: 0.5 g plastics, 0.2 g catalyst, input power of ca. 56 W, 30 mL/min, H_2:Ar=3:7, 1 h, without external heating, ambient pressure.).

4. Conclusion

In this study, polyolefin wastes were effectively converted to liquid fuels over the Sm/USY catalysts through a plasma-assisted catalysis system without external heating. By optimizing the reaction parameters, LDPE plastic was fully converted over the optimized 5Sm/USY catalyst under plasma condition, achieving 83 % yield of liquid product with a 98 % selectivity of gasoline-range components. The modification of Sm species into USY zeolite improved the amount of L acid sites, leading to an enhanced C-H bonds activity ability. More importantly, the proximity coupling of L acid sites and B acid sites over the 5Sm/USY catalyst effectively cleave the C-H and C-C bonds in polyolefins, and promoted the isomerization-hydrogenation progress. Meanwhile, the shape-selective effect of porous zeolite support facilitated to the formation of gasoline-range components with high selectivity in liquid product. Furthermore, the energy cost under plasma-catalytic mode was significantly lower than that under thermal-catalytic mode, giving prominence to the advantage of plasma catalysis. Additionally, the 5Sm/USY catalyst showed a promising applicability to plasma-assisted recycling of mixed polyolefin plastics and commercial plastics. Therefore, this study provided a potential route with energy-saving benefit for plasma-catalytic polyolefin wastes to value-added chemicals under relatively mild operation.

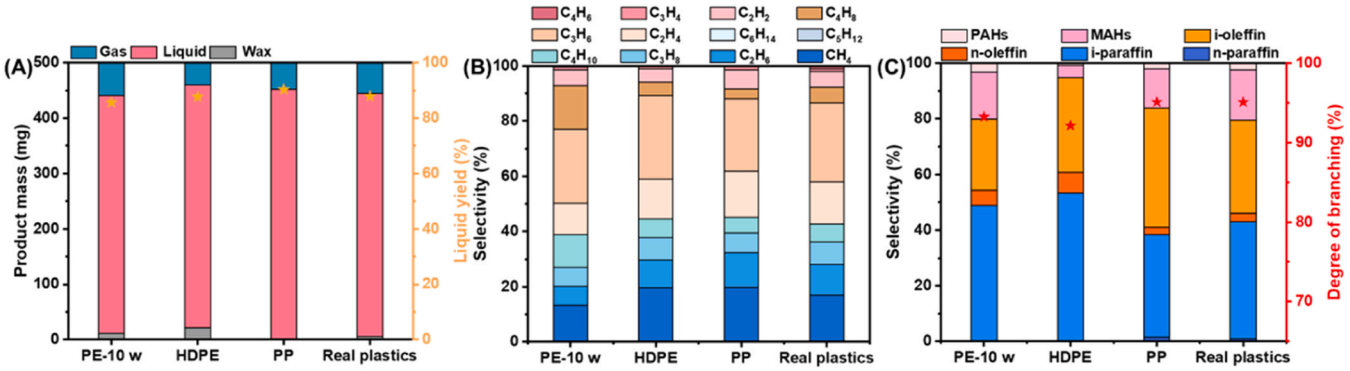


Fig. 9. Plasma-catalytic performance of different plastics over 5Sm/USY catalyst: (A) product mass, (B) the selectivity of different components in gaseous product, (C) the selectivity of different components in liquid product. (Reaction conditions: 0.5 g plastics, 0.2 g catalyst, input power of ca. 56 W, 30 mL/min, H_2:Ar=3:7, 1 h, without external heating, ambient pressure.).

CRediT authorship contribution statement

Tianhao Shi: Data curation. **Jianhui Han:** Writing – review & editing, Data curation. **Bingbing Chen:** Writing – review & editing, Data curation. **Tianqi Yun:** Writing – review & editing, Writing – original draft, Methodology, Data curation. **Chuan Shi:** Writing – review & editing, Supervision, Project administration, Methodology, Funding acquisition, Conceptualization. **Yanan Diao:** Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Data curation. **Chengxin Hou:** Writing – original draft, Software, Methodology, Investigation, Data curation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.apcatb.2025.125944.

Data availability

No data was used for the research described in the article.

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